Shubh Lavti

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LinkedIn | Github

Education

Indian Institute of Technology, Gandhinagar *2019 - 2023*

- Bachelor of Technology in Computer Science and Engineering **CGPA: 8.20/10**

Technical Skills

Core Competencies: Algorithms, Data Structures, Machine Learning, Data Science and Data Engineering
Programming Languages: JAVA, Python3, C/C++, MATLAB, SQL
Tools : AWS, Docker, Pandas, PySpark, SQL, MariaDB, YugaByteDB, Tensorflow, PyTorch, Perforce,

Professional Experience

Software Engineer: *Jan 2024 - Present*
[Rakuten Mobile Inc.] Tokyo, Japan

- Designed and deployed a scalable **PySpark** pipeline for daily ETL processing of large datasets from MinIO, performing aggregations to meet business requirements.
- Stored the aggregated data in YugabyteDB, enabling faster analytics and reducing ad hoc task time by 2 hours, enhancing overall productivity.
- Developed an interactive Web Portal using Python and FastAPI, leveraging LLM to extract meaningful insights from telecom data cosisting of 3000+ files using **RAG**. Facilitated improved decision-making for network and business teams.

Software Development Engineering Intern: *June 2023 - Oct 2023*
[Sprih Labs Pvt Ltd.] Pune, India

- Developed and maintained REST APIs and Backend features using Java, applying strong OOP principles and **MVC architecture**. Used MySQL database to ensure data integrity and performance.
- Developed an automated backend feature for bill categorization using message broker technology, ensuring efficient data processing, storage, and real-time system updates.
- Developed key backend features for supply chain management, integrating vendor and tenant data while implementing complex calculations to ensure seamless operations and efficiency.

Engineering Development Group Intern: *May 2022 - July 2022*
[Mathworks Inc.] Bengaluru, India

- Enhanced internal MATLAB functions, including the **getFrame** command for supporting multiple object types, and developed the **copygraphics** function for multi-mime type storage on the system clipboard, gaining expertise in C++ and OOP.

Projects

Covid-19 Data Analysis [Prof. Anirban Das Gupta | IIT Gandhinagar] *April 2022 - May 2022*

- Analyzed growth patterns and implemented ARIMA analysis to forecast future COVID numbers.
- Achieved forecast accuracy with a Mean Absolute Percentage Error (MAPE) of less than 5%, providing reliable insights for decision making.

Library Database Management System [Prof. Mayank Singh | IIT Gandhinagar] *Jan 2022 - April 2022*

- Designed an institute-based library management system from scratch. Supported features such as book transactions, receipt generation, accounting for faculty hours, and asset information per employee.
- Implemented the management system using MySQL and integrated with HTML using a python based web-framework, Flask.

Emoji Suggestor [Prof. Mayank Singh | IIT Gandhinagar] *Aug 2021 - Nov 2021*

- Proposed and implemented a model combining a pre-trained Transformer (BERT) cascaded with a Multi-Layer Bi-Directional LSTM to predict appropriate emoji within text as intended.
- The proposed model gave a high accuracy score of up to 97.6%. Poster

Relevant Courses

Data Structures and Algorithms, Operating Systems, Compilers, Object Oriented Programming, Databases, Computing, Natural Language Processing, Machine Learning, Introduction to Data Science, Digital Systems, Discrete Mathematics, Theory of Computation, Computer Organization and Architecture